

Quiz 2

Math 213

Name:

Student Number:

1. Determine $|V|$ for G such that G has nine edges and all vertices have degree 3.
2. Consider a graph $G = (V, E)$ of n vertices and m edges.
Let v be a vertex and e an edge of G . Give the order(number of vertices) and the size(number of edges) of G^c , $G - v$ and $G - e$.
3. Find out whether the complement of a regular graph is regular. If so, prove it; if not, give a counterexample.

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1. Determine $|V|$ for G such that G is regular with 15 edges.
2. Give an example of a connected graph G where removing any edge of G results in a disconnected graph.
3. Let $G = (V, E)$ be a graph and v a vertex of G . prove that:
If G is not connected, then G^c is connected

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1. Determine $|V|$ for G such that G has 10 edges with two vertices of degree 4 and all others of degree 3.

2. If $G = (V, E)$ is an undirected graph with $|V| = v$, $|E| = e$ and no loops, prove that $2e < v^2 - v$.

3. Let $m, n \in \mathbb{Z}^+$ with $m < n$. How many paths of length m are there in the complete graph K_n ?

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1. Give the number of edges of an r -regular graph with n vertices.

2. If G is a self-complementary graph on n vertices, where $n > 1$, prove that $n = 4k$ or $n = 4k + 1$, for some $k \in \mathbb{Z}^+$.

3. Which of the following graphs are isomorphic (Explain why).

